Fiscal Year:	FY 2004	**Task Last Updated:**	FY 03/29/2006
PI Name:	Ngo, Que Ph.D.		
Project Title:	Effect of Hypobaria on Genetic Expression of Ethylene, Gas Exchange, and Crop Productivity		
Division Name:	Human Research		
Program/Discipline:	ADVANCED HUMAN SUPPORT TECHNOLOGIES		
Program/Discipline--Element/Subdiscipline:	ADVANCED HUMAN SUPPORT TECHNOLOGIES--Advanced life support		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	77843-2133	**Congressional District:**	31
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	GSRP
Start Date:	10/01/2003	**End Date:**	10/01/2004
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA HQ
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:			
Performance Goal No.:			
Performance Goal Text:			
Task Description:	The development of an Advanced Life Support (ALS) System will incorporate air and water revitalization and food production as roles for a plant growth system. Ethylene accumulation to levels detrimental to plant growth is problematic for space shuttles, the international space station, and potentially in Lunar/Martian plant growth facilities. Low pressure (hypobaria) reduces ethylene production. The proposed research objectives are 1) to characterize plant gas exchange and growth under low pressures which applies to an ALS System, and 2) to use molecular biology in Arabidopsis thaliana to determine how ethylene biosynthetic enzymes are affected genetically under hypobaria. Long-term goals involve the application of this research to NASA baseline crops (i.e. lettuce) to improve the NASA ALS System.		
Rationale for HRP Directed Research:			

Research Impact/Earth Benefits:	Significant findings from the proposed research will increase the feasibility of using the Arabidopsis plant system as a model for the development of NASA baseline crops such as lettuce or wheat. By implementing molecular biology, it would be possible to identify the specific genes involved in ethylene biosynthesis and apply this knowledge for the growth and development of NASA baseline crops. The application of this plant system to NASA baseline crops is of great importance for the improvement of NASA's ALS System in future endeavors to support life in space.
Task Progress:	No progress report this period.
Bibliography Type:	Description: (Last Updated: 02/27/2004)